COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 9 : SLO 2 — Restoring Division (Simulation & Analysis)
Name: JATANPREET SINGH VIRDI JATANPREET SINGH VIRDI | Reg. No: RA2511003011374 | Solved Worksheet

Section C: Paper Simulation (Apply): 0110 (6) ÷ 0010 (2) — restoring division

• Init: A = 0000, Q = 0110, M = 0010

• Step 1: shift (A,Q) → A = 0000, Q = 1100; A = A - M = 1110 (< 0) → restore A = 0000, Q0 = 0 → A = 0000, Q = 1100

• Step 2: shift → A = 0001, Q = 1000; A = A - M = 1111 (< 0) → restore A = 0001, Q0 = 0 → A = 0001, Q = 1000

• Step 3: shift → A = 0011, Q = 0000; A = A - M = 0001 (≥ 0) → Q0 = 1 → A = 0001, Q = 0001

• Step 4: shift → A = 0010, Q = 0010; A = A - M = 0000 (≥ 0) → Q0 = 1 → A = 0000, Q = 0011

• Final Quotient = 0011 (3), Remainder = 0000 (0) (6 ÷ 2 = 3 remainder 0 ✓)

Section D: Role-Based Play Debrief (Analyze)

• Restoration is triggered whenever the subtraction produces a negative remainder — the subtraction 'failed', so the divisor is added back to recover the previous remainder.

• The division is complete after n cycles, where n is the number of bits of the quotient/dividend register; the quotient sits in Q and the remainder in A.

• Each left shift brings the next dividend bit into the remainder for the trial subtraction and simultaneously frees the LSB of Q to receive the new quotient bit.